

Big Data
Progress Mingguan



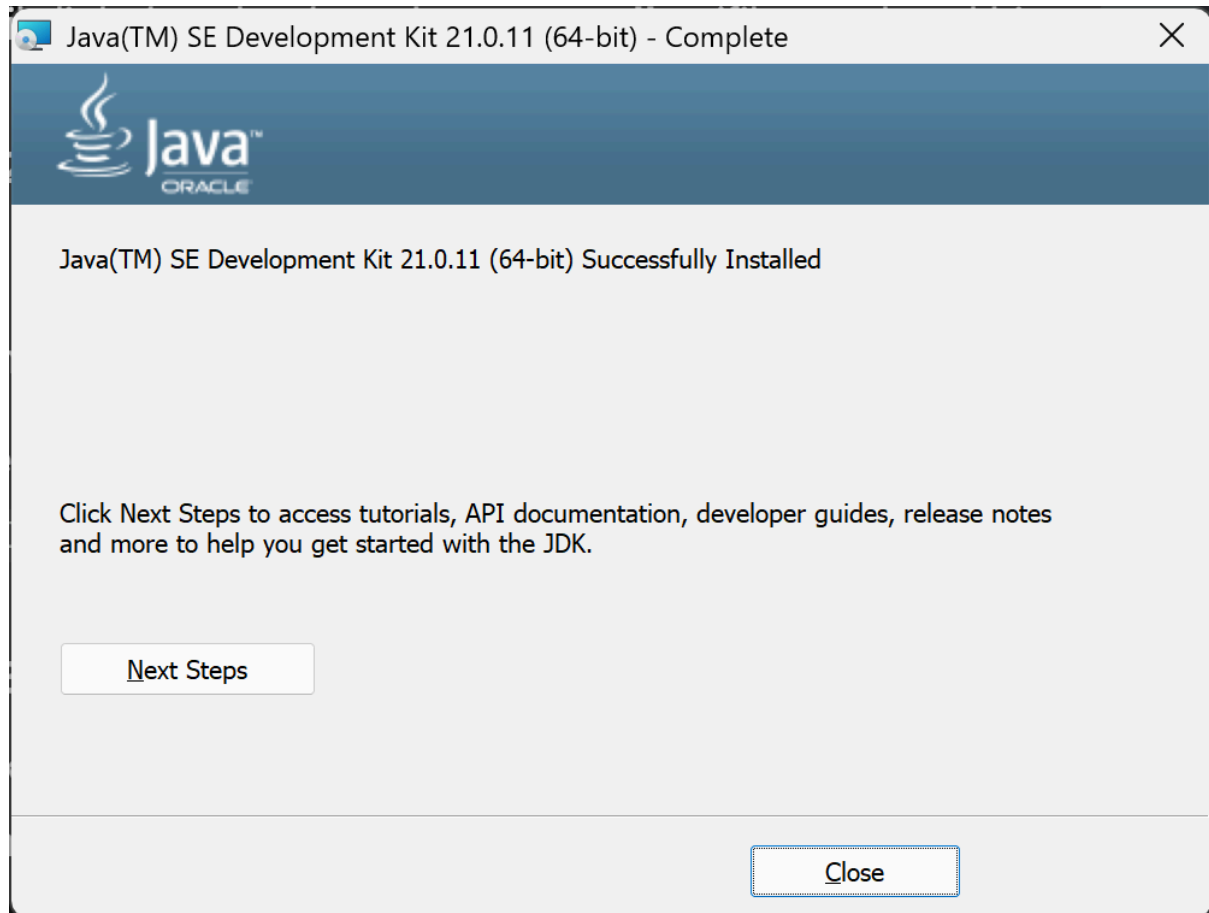
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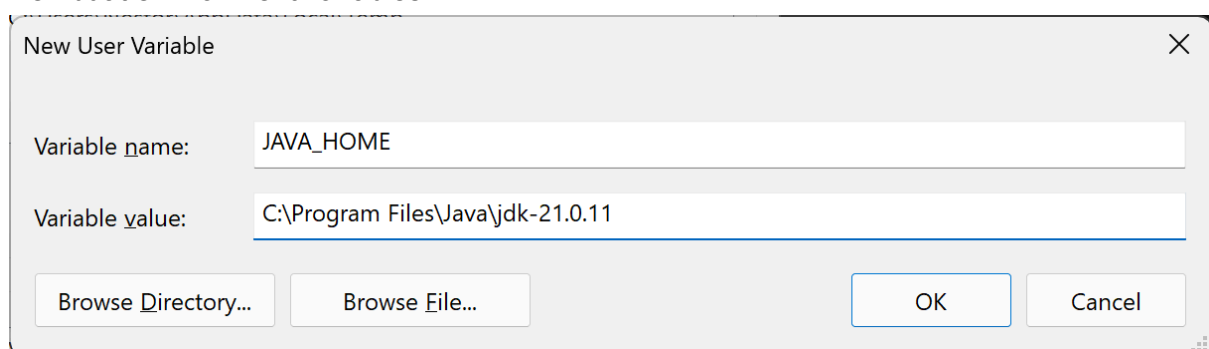
PROGRAM STUDI D-IV TEKNOLOGI REKAYASA INTERNET
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Install JDK 21



Screenshot ini menunjukkan proses instalasi Java Development Kit (JDK) versi 21.0.11 (64-bit) telah berhasil diselesaikan di Windows. Instalasi Java ini diperlukan sebagai syarat utama sebelum menginstal Apache Spark, karena sistem Spark berjalan menggunakan Java Virtual Machine (JVM).

Membuat environment variables



Edit environment variable



C:\Program Files\Common Files\Oracle\Java\javapath
%SystemRoot%\system32
%SystemRoot%
%SystemRoot%\System32\Wbem
%SYSTEMROOT%\System32\WindowsPowerShell\v1.0\
%SYSTEMROOT%\System32\OpenSSH\
C:\Program Files (x86)\NVIDIA Corporation\PhysX\Common
C:\Program Files\dotnet\
C:\Program Files\NVIDIA Corporation\NVIDIA app\NvDLISR
C:\Users\Nestor\Downloads\mongodb-database-tools-windows-x...
C:\Users\Nestor\Downloads\mongodb-database-tools-windows-x...
%JAVA_HOME%\bin

New

Edit

Browse...

Delete

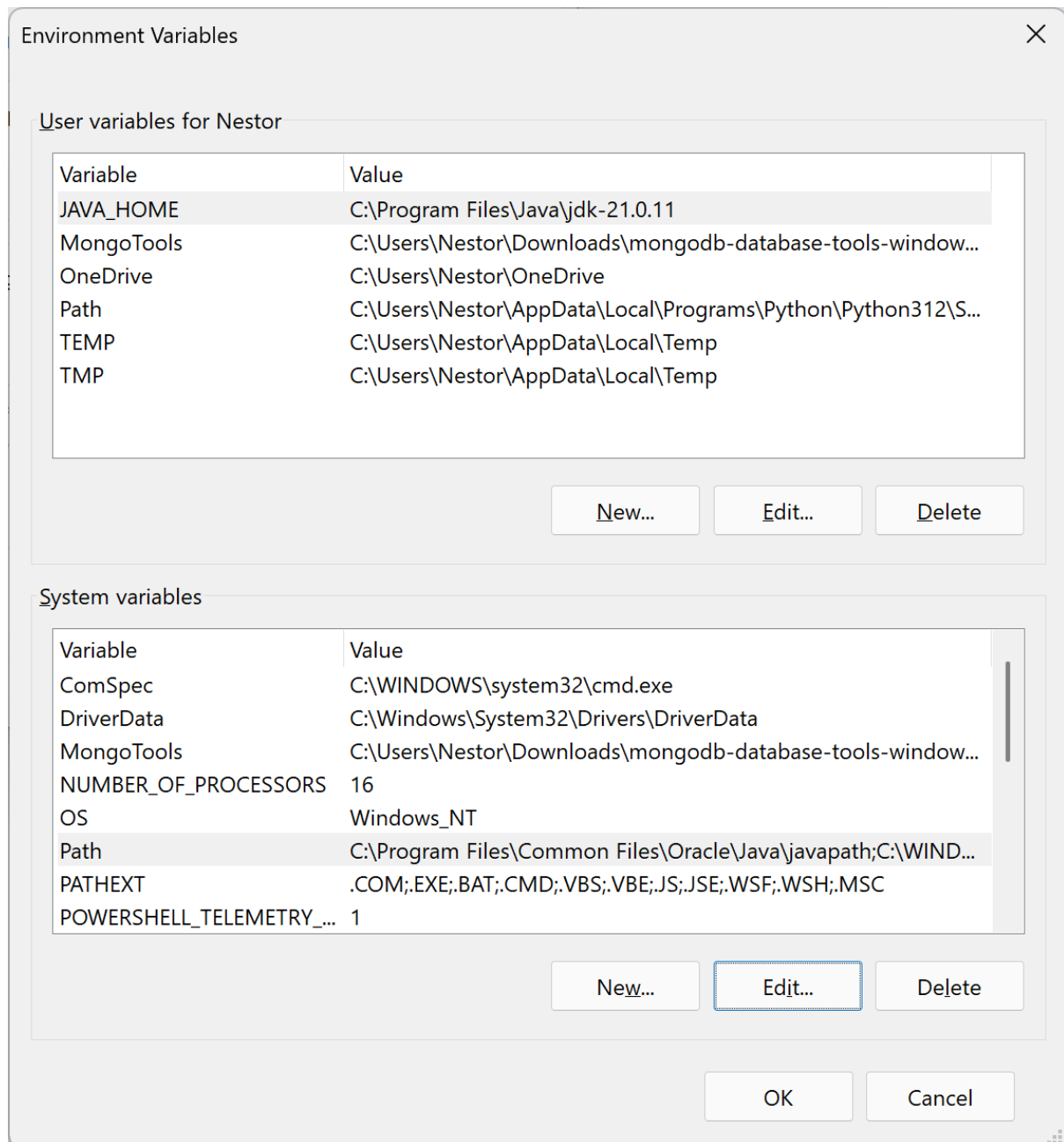
Move Up

Move Down

Edit text...

OK

Cancel



Konfigurasi Environment Variables dilakukan dengan membuat variabel baru bernama JAVA_HOME yang diarahkan ke folder instalasi Java, yaitu C:\Program Files\Java\jdk-21.0.11. Selain itu, %JAVA_HOME%\bin juga ditambahkan ke dalam variabel path di sistem. Hal ini dilakukan agar command Java bisa dijalankan langsung dari Command Prompt (CMD).

Java terinstall

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nestor>java -version
java version "21.0.11" 2026-04-21 LTS
Java(TM) SE Runtime Environment (build 21.0.11+9-LTS-211)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.11+9-LTS-211, mixed mode, sharing)

C:\Users\Nestor>
```

Command `java -version` dijalankan pada Command Prompt untuk memastikan Java sudah terinstal dan terbaca oleh sistem. Output menampilkan `java version "21.0.11"`, yang berarti Java sudah siap digunakan untuk menjalankan aplikasi Big Data.

Install pyspark

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nestor>python --version
Python 3.12.6

C:\Users\Nestor>pip install pyspark
Collecting pyspark
  Downloading pyspark-4.1.1.tar.gz (455.4 MB)
    455.4/455.4 MB 3.3 MB/s 0:05:46
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting py4j<0.10.9.10,>=0.10.9.7 (from pyspark)
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl.metadata (1.3 kB)
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl (203 kB)
Building wheels for collected packages: pyspark
  Building wheel for pyspark (pyproject.toml) ... done
  Created wheel for pyspark: filename=pyspark-4.1.1-py2.py3-none-any.whl size=456008747 sha256=e6bb4c67e68ddea6889a6f4357f3167f4f59e35f939cda9d5d8dc48c94321182
  Stored in directory: c:\users\nestor\appdata\local\pip\cache\wheels\4\ca\ea\203f40b3e935bbf99bee851c2f4a87d22996ab8212d367ce58
Successfully built pyspark
Installing collected packages: py4j, pyspark
Successfully installed py4j-0.10.9.9 pyspark-4.1.1

[notice] A new release of pip is available: 26.0.1 -> 26.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\Nestor>
```

Proses instalasi Apache Spark dilakukan menggunakan command `pip install pyspark`. Pada tahap ini, sistem mengunduh dan menginstal *library* PySpark versi 4.1.1 beserta kebutuhan lainnya ke dalam Python 3.12.6 yang sudah ada di laptop.

Spark berjalan

```
Command Prompt - pyspark
C:\Users\Nestor>set PYSPARK_PYTHON=python
C:\Users\Nestor>set PYSPARK_DRIVER_PYTHON=python
C:\Users\Nestor>echo %JAVA_HOME%
C:\Program Files\Java\jdk-21.0.11

C:\Users\Nestor>pyspark
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
WARNING: Using incubator modules: jdk.incubator.vector
26/04/29 20:29:27 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: H
ADOOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/confluence/display/HADOOP2/WindowsProblems
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/04/29 20:29:31 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
Welcome to

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version 4.1.1

Using Python version 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024 20:11:23)
Spark context Web UI available at http://TOR-101:4040
Spark context available as 'sc' (master = local[*], app id = local-1777469372558).
SparkSession available as 'spark'.
>>>
```

Command set PYSPARK_PYTHON=python dan set PYSPARK_DRIVER_PYTHON=python digunakan untuk mengarahkan Spark agar membaca *path* Python yang benar di Windows. Setelah itu, command pyspark dijalankan dan berhasil memunculkan antarmuka Spark versi 4.1.1. Ini membuktikan bahwa *engine* Apache Spark sudah berjalan dan siap digunakan untuk memproses data dari IoT.